



**UNIVERSITY OF SCIENCE
HO CHI MINH CITY**

Intellectual Property

Adapted from Quinn's Slides

Nguyen V. Vu

Summer 2026

Introduction

- Value of intellectual properties much greater than value of media
 - Difficult to create, easy to copy
- Illegal copying pervasive
- How should we treat intellectual property?



What Is Intellectual Property?

- Intellectual property: any unique product of the human intellect that has **commercial** value



- Intellectual property $\neq$ physical manifestation

Property Rights

- Locke: *The Second Treatise of Government*
- People have a right...
 - ❑ to property in their own person
 - ❑ to their own labor
 - ❑ to things which they remove from Nature through their labor
- As long as...
 - ❑ no person claims more property than he or she can use
 - ❑ after someone removes something from common state, there is plenty left over



Expanding the Argument to Intellectual Property

- Writing a play akin to making a belt buckle
- Belt buckle
 - Mine ore
 - Smelt it down
 - Cast it
- Writing a play
 - “Mine” words from English language
 - “Smelt” them into prose
 - “Cast” them into a complete play



Analogy Is Imperfect

- If two people write same play, both cannot own it
 - every intellectual property is one-of-a-kind
- If one person “takes” another’s play, both have it
 - copying an intellectual property is different from stealing a physical object

Benefits of Intellectual Property Protection

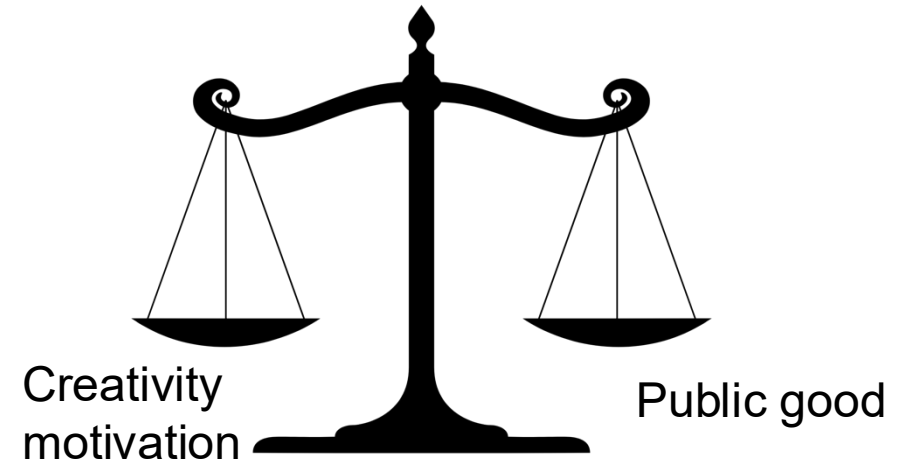
- Some people are altruistic; some are not
- Allure of wealth can be an incentive for speculative work



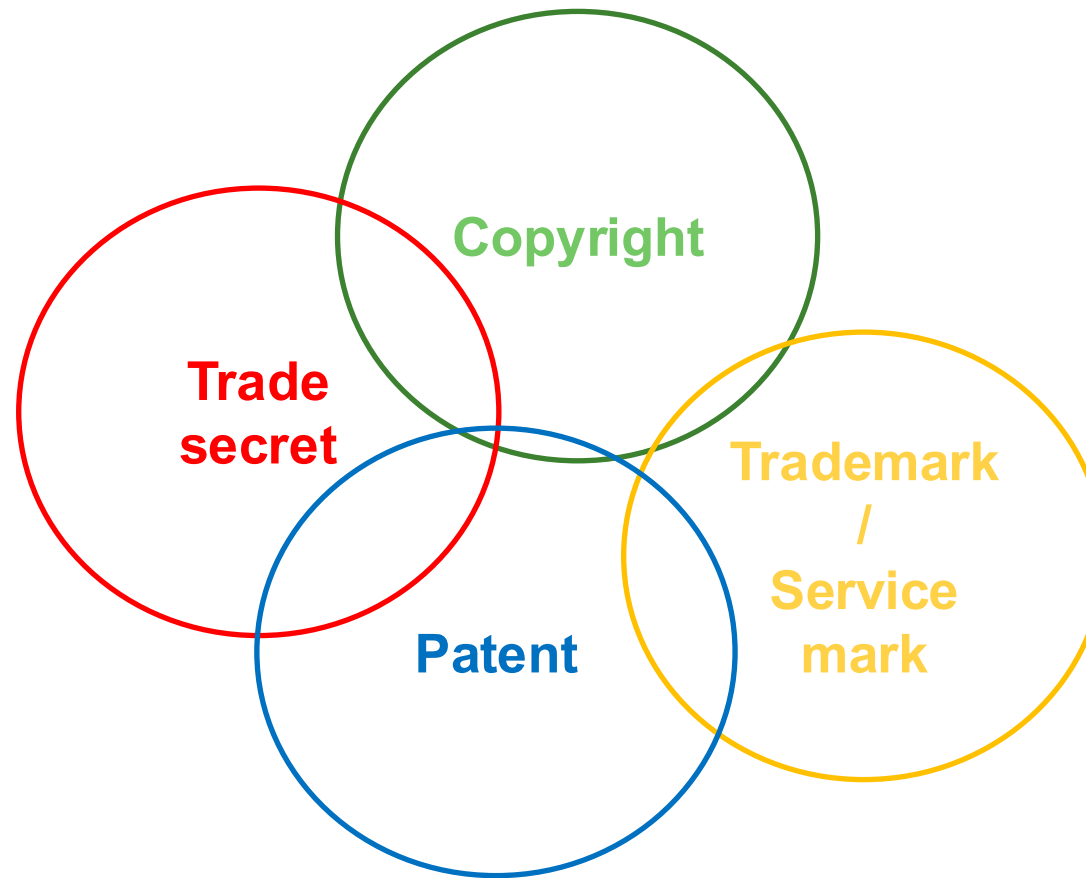
Credit: Shutterstock

Limits to Intellectual Property Protection

- Giving creators rights to their inventions stimulates creativity
- Society benefits most when inventions in public domain

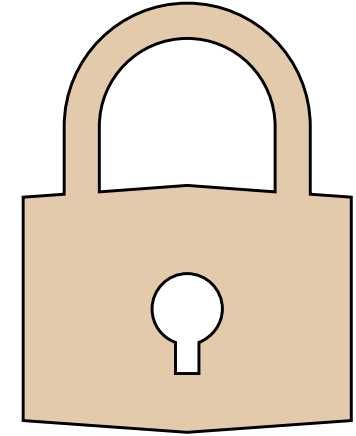


Protecting Intellectual Property



Trade Secret

- Confidential piece of intellectual property that gives company a competitive advantage
 - E.g., formulas, processes, designs, strategic plans, customer lists
- Never expires
- Not appropriate for all intellectual properties
- Reverse engineering allowed
- May be compromised when employees leave firm



Trademark, Service Mark

- Trademark: identifies goods
- Service mark: identifies services
- Company can establish a “brand name”
- Does not expire
- Companies advertise to protect their trademarks
- Companies also protect trademarks by contacting those who misuse them



Patent

- A public document that provides detailed description of invention
- Provides owner with exclusive right to the invention
- Owner can prevent others from making, using, or selling invention for 20 years (US Patent)



Copyright

- Provides owner of an original work five rights
 - ❑ Reproduction
 - ❑ Distribution
 - ❑ Public display
 - ❑ Public performance
 - ❑ Production of derivative works



Fair Use Concept

- Sometimes **legal** to reproduce a copyrighted work without permission
- Courts consider four factors
 - ❑ Purpose and character of use
 - ❑ Nature of work
 - ❑ Amount of work being copied
 - ❑ Affect on market for work



[This Photo](#) by Unknown Author is licensed under [CC BY-SA-NC](#)

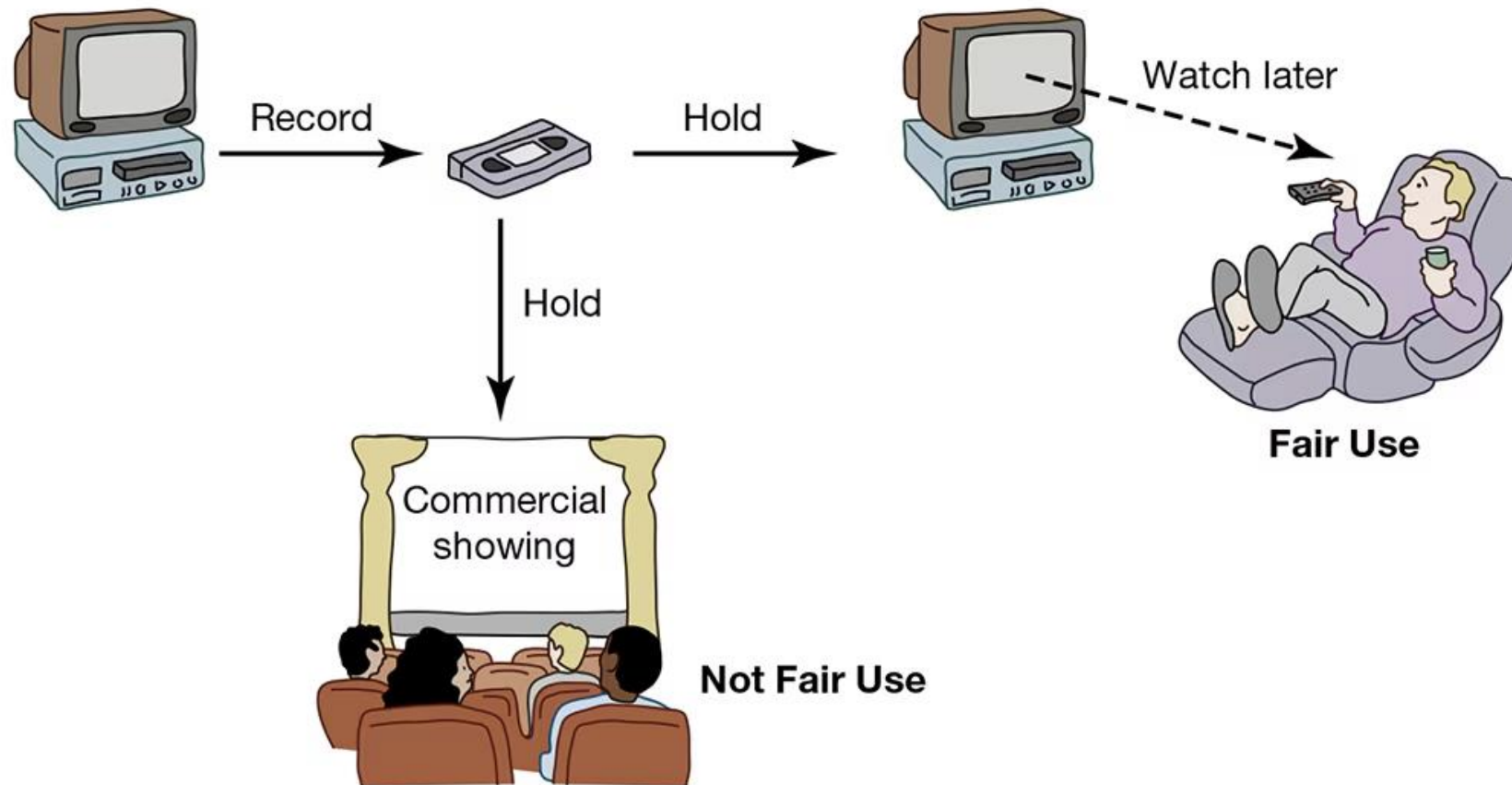
Discussion

- You love a live show on TV. One day you are busy to watch it live so you use a recording device (VCR) to record and watch it later.
- Is your action legal?
- Does the VCR manufacturer (like Sony) violate the copyright law (infringement)?



[This Photo](#) by Unknown Author is licensed under [CC BY-SA-NC](#)

Discussion – U.S. Supreme Court's ruling



Digital Millennium Copyright Act (DMCA)

- First big revision of U.S. copyright law since 1976
- Made it illegal for anyone to
 - ❑ Circumvent encryption schemes placed on digital media
 - ❑ Circumvent copy controls, even for fair use purposes
 - ❑ Sell software designed to circumvent such schemes
 - ❑ Discuss circumvention online



This Photo by Unknown Author is
licensed under [CC BY-NC-ND](#)

Vietnam's Intellectual Property Law

■ Intellectual Property Law

- ❑ Effective since 1 July, 2006
- ❑ Covering intellectual properties like artworks, patents, trade secrets, trademarks/service marks, seeds and seed reproduction equipment

■ Protection duration

- ❑ Trademarks, service marks, inventions: 20 years
 - ❑ Certain inventions aka “useful solutions” : 10 years
- ❑ Artworks: 50 years
- ❑ Other works: lifetime

Digital Rights Management

- Actions owners of intellectual property take to protect their rights
- Approaches
 - Encrypt digital content
 - Digital marking so devices can recognize content as copy-protected

Peer-to-Peer Networks

- Peer-to-peer network
 - ❑ Transient network
 - ❑ Connects computers running same networking program
 - ❑ Computers can access files stored on each other's hard drives

- How P2P networks facilitate data exchange
 - ❑ Give each user access to data stored in many other computers
 - ❑ Support simultaneous file transfers among arbitrary pairs of computers
 - ❑ Allow users to identify systems with faster file exchange speeds

BitTorrent

- Broadband connections: download much faster than upload
- BitTorrent speeds downloading
 - Files broken into pieces
 - Different pieces downloaded from different computers
- Used for downloading large files
 - Computer programs
 - Television shows
 - Movies

Software Copyrights

- Copyright protection began 1964
- What gets copyrighted?
 - Expression of idea, not idea itself
 - Object program, not source program
- Companies treat source code as a trade secret

Software Patents

Question	Software Copyright	Software Patent
What is protected?	Object program, screen displays	Software process with practical utility
Is getting protection expensive?	No	Yes
Is getting protection time consuming?	No	Yes
Is reverse engineering allowed?	Yes	No

Violations of Software Copyrights

- Copying a program to give or sell to someone else
- Preloading a program onto the hard disk of a computer being sold
- Distributing a program over the Internet

Safe Software Development

- Reverse engineering okay
- Companies must protect against unconscious copying
- Solution: “clean room” software development strategy
 - Team 1 analyzes competitor’s program and writes specification
 - Team 2 uses specification to develop software

Open-Source Software

- Consequences of proprietary software
- Open source definition
- Beneficial consequences of open-source software
- Examples of open-source software
- The GNU Project and Linux
- Impact of open-source software
- Critique of the open-source software movement

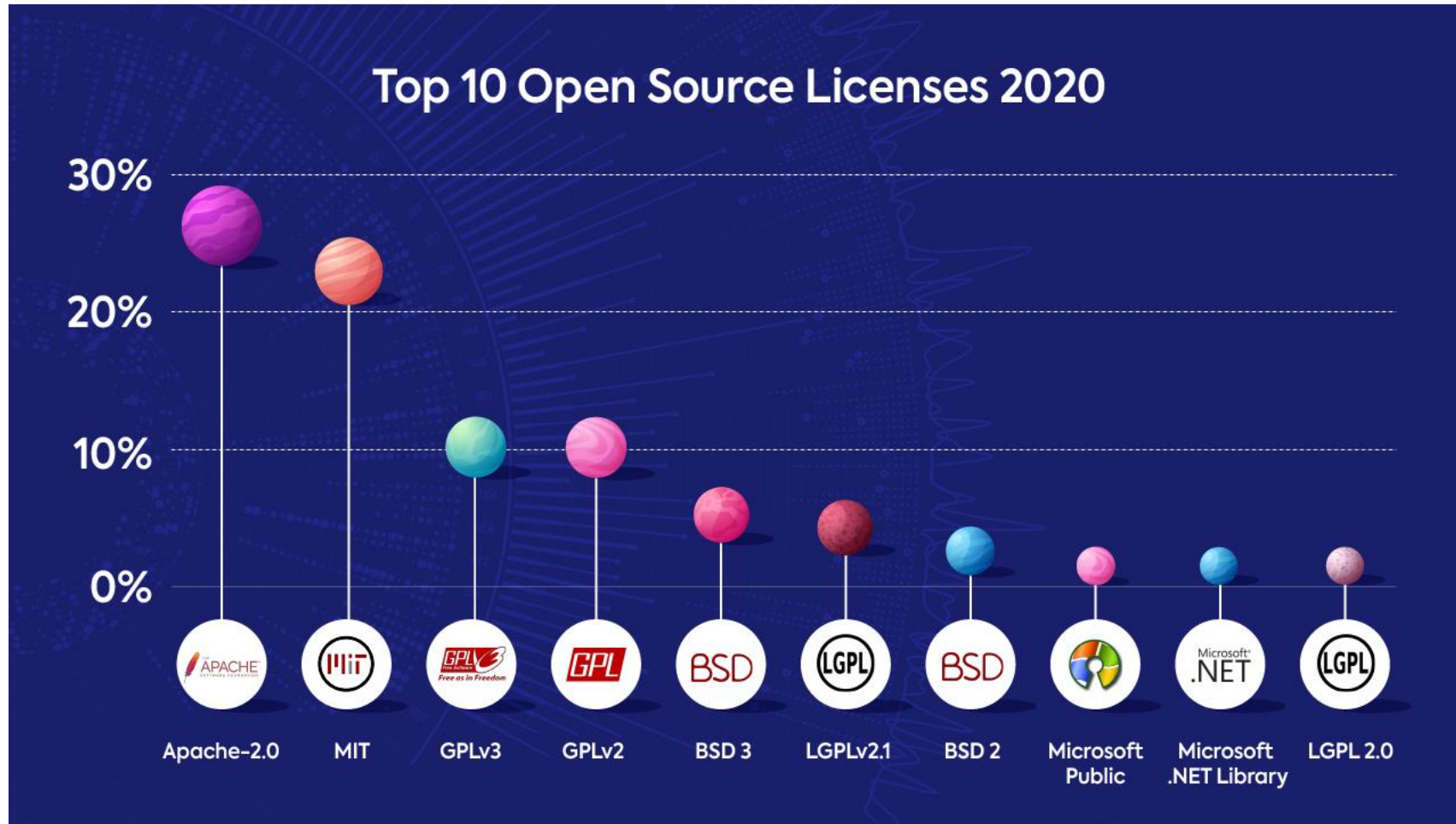
Consequences of Proprietary Software

- Increasingly harsh measures being taken to enforce copyrights
- Copyrights are not serving their purpose of promoting progress
- It is wrong to allow someone to “own” a piece of intellectual property

Open Source Definition

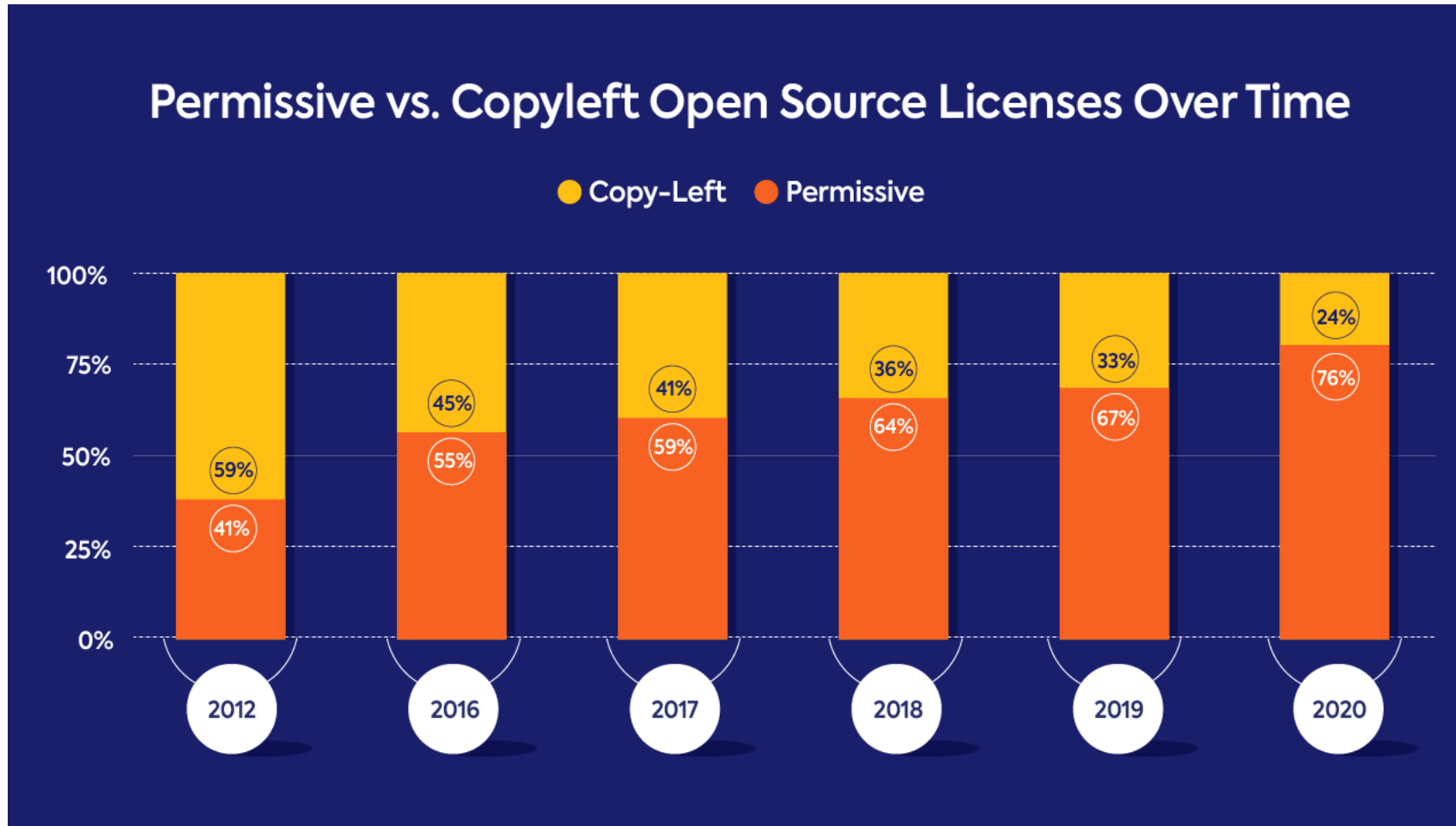
- Source code included in distribution
- No restrictions preventing others from modifying source code
- No restrictions regarding how people can use software
- Same rights apply to everyone receiving redistributions of the software (copyleft)

Open Source Licenses



Source: <https://www.whitesourcesoftware.com/resources/blog/open-source-licenses-trends-and-predictions/>

Open Source Licenses Restriction Trends



Source: <https://www.whitesourcesoftware.com/resources/blog/open-source-licenses-trends-and-predictions/>

Discussion

- Debate the benefits and harms of open-source software
- Are you against or for it? Why?
- Let's divide into two groups against and for to debate

Benefits of Open-Source Software

- Gives everyone opportunity to improve program
- New versions of programs appear more frequently
- Eliminates tension between obeying law and helping others
- Programs belong to entire community
- Shifts focus from manufacturing to service

Impact of Open-Source Software

- Linux putting pressure on
 - companies selling proprietary versions of Unix
 - Microsoft and Apple desktops

Critique of the Open-Source Software Movement

- Without critical mass of developers, quality can be poor
- Installation issues
- Without an “owner,” incompatible versions may arise
- Relatively weak GUI
- Poor mechanism for stimulating innovation
 - no companies will spend billions on new programs

Legitimacy of Intellectual Property Protection for Software

- Software licenses typically prevent you from making copies of software to sell or give away
- Software licenses are legal agreements

Rights-based Analysis

- “Just deserts” argument
 - Programming is hard work that only a few can do
 - Programmers should be rewarded for their labor
 - They ought to be able to own their programs

- Criticism of “just deserts” argument
 - Why does labor imply ownership?
 - Can imagine a just society in which all labor went to common good
 - Intellectual property not like physical property

Utilitarian Analysis

- Argument against copying
 - ❑ Copying software reduces software purchases...
 - ❑ Leading to less income for software makers...
 - ❑ Leading to lower production of new software...
 - ❑ Leading to fewer benefits to society
- Each of these claims can be debated
 - ❑ Not all who get free copies can afford to buy software
 - ❑ Open-source movement demonstrates many people are willing to donate their software-writing skills
 - ❑ Hardware industry wants to stimulate software industry
 - ❑ Difficult to quantify how much society would be harmed if certain software packages not released

Creative Commons



- Problems with the current law
 - Eligible works are copyrighted the moment they are created
 - No copyright notice does not mean it's okay to copy
 - Must contact people before using work
- That slows down creative re-use of others' work
- Free Creative Commons license indicates
 - Which kinds of copying are okay
 - Which rights are being retained
- Flickr, Magnatune, and MIT OpenCourseWare are well-known sites using Creative Commons licenses

Debate

A plane makes an emergency crash landing on a deserted tropical island. Two dozen survivors must fend for themselves until help arrives. All of them are from large cities and none of them has camping experience. The survivors find it impossible to gather enough food, and everyone begins losing weight. One person spends a lot of time alone and figures out how to catch fish. They bring fish back to the camp. Others ask them to teach them how to catch fish. They refuse but offer to share the fish they have caught with the other passengers as long as they take care of the other camp chores, such as hauling fresh water, gathering firewood, and cooking.

Task: Four members each group. Half of the groups discuss and come up with reasons why fisherman's position is **morally wrong**. The other groups find reasons why his position is **acceptable**.

Discussion

- Who owns a song created by AI?
 - <https://www.nytimes.com/2023/04/15/business/dealbook/artificial-intelligence-copyright.html>